higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(First Paper)
NQF LEVEL 3

20 FEBRUARY 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from a table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitute into formula
R/J	Reasoning/Justification
P	Penalty – no units, incorrect rounding off
R	Rounding off
E	Explanation
U	Unit
MF	Manipulation of formula

This marking guideline consists of 9 pages.

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QUESTION 1 [30] *Do not penalise if units are omitted unless marks are allocated.			
Question	**Solution**		**Explanation**
1.1 1.1.1	$8 + (3{,}6 + 3{,}94) \times 2 \div (4 + 3)$ $= 8 + 7$✓ $\times 2 \div 7$✓ BODMAS $= 8 + 14$✓$\div 7$ $= 8 + 2$ $= 10$✓ (**answer only 1 mark**)		1MA + 7 1MA ÷ 7 1MA + 14 1A 10 (4)
1.1.2	$\sqrt{\frac{64}{16}}$ $= \frac{8}{4}$✓ $= 2$✓ BODMAS (**answer only 1 mark**)		1MA $\frac{8}{4}$ (no mark if only one correct) 1A 2 (2)
1.2 1.2.1	Fraction = $\frac{7}{28}$✓$= \frac{1}{4}$✓ (**answer only full marks**)		1MA 1S (2)
1.2.2	Fraction = $\frac{36}{3600}$✓$= \frac{1}{100}$✓ (**answer only full marks**)		1MA 1S (2)
1.2.3	Fraction = $\frac{25}{100}$✓$= \frac{1}{4}$✓ (**answer only full marks**)		1MA 1S (2)
1.3 1.3.1	Time to fill reservoir = 27 000 ÷ 900✓ = 30 hours✓ = 1 day and 6 hours✓		1M 1A 1C (3)
1.3.2	$18 \times 900 = 16\,200l$ ✓ $= \frac{16200}{27000}$✓$\times 100\%$✓ $= 60\%$✓		1MA 16 200l 1M $\frac{16200}{27000}$ 1M $\times 100\%$ 1A 60 (4)
1.4.1	Thabo's rating = 22 ÷ 25 = 0,88 and Esther's rating = 43 ÷ 50 = 0,86✓ Thabo received the best rating✓	Thabo's rating = 88% and Esther's rating = 86%✓ Thabo received the best rating✓ Or	1MA (no mark if only one correct) 1A

	Or	Thabo's rating = $\frac{44}{50}$ ✓ Thabo received the best rating✓	(2)
1.4.2	44✓ : 43✓		2A (2)
1.5	Subtract the times: Now borrow 1 minute: Difference in time = 15 minutes✓ and 52 seconds✓ **Accept alternate methods** (**answer only full marks**)		1M 2A (3)
1.6	Reading Accept any answer between 181 °*C* to 184 °*C*✓✓		2 Reading from thermometer (2)
1.7	°F = $200 \times \frac{9}{5} + 32$✓ °F = 392✓ (**answer only full marks**)		1SF 1A (2)
			[30]

	1	:39	:04
-	1	:23	:12

	1	:38	:64 ✓
-	1	:23	:12
=		:15	:52

QUESTION 2 [30] *Do not penalise if units are omitted unless marks are allocated.		
Question	**Solution**	**Explanation**
2.1 2.1.1	Koppie✓ and Shower✓	2RG (2)
2.1.2	A5;B5✓ or A5 and B5✓	1RG both coordinates must be correct (1)
2.1.3	South East✓	1A (1)
2.1.4	$PQ^2 = 600^2 + 600^2$ ✓✓ PQ = $\sqrt{600^2 + 600^2}$ ✓ PQ = 848,528 ✓m PQ = 849m	1RG and 1SF 1 MF 1A 848,528 (4)
2.1.5	Perimeter	

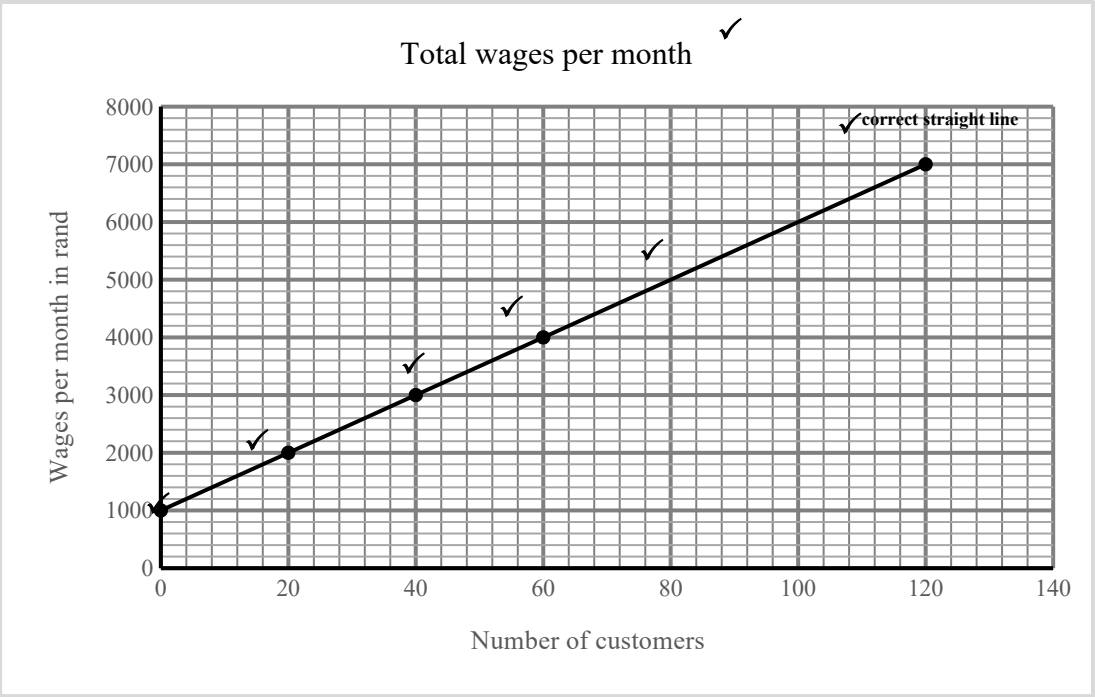
	= 849✓ + 1050✓+ 900✓ + 150✓ +900+ 1800 + 1200 ✓ - 10✓ = 6 839 ✓m = 6,839 ✓km Accept (6,8 and 6,84)	1A Reading from Q2.1.4 (849) 2RG 1050 and 150 2RG all other measurements 1M -10 1 A 6 839 1C (from m to km) (8)
2.2.1	Square✓	1A (1)
2.2.2	Area = 7^2✓ Area = 49 ✓cm^2 (**answer only full marks**)	1SF 1A (2)
2.2.3	Volume = 49 ✓× 14✓ Volume = 686 cm^3✓ Or Volume = 7✓× 7 × 14✓ Volume = 686 cm^3✓ (**answer only full marks**)	2SF 1CA with correct unit 2SF 1CA with correct unit (3)
2.2.4	Volume = 686 cm^3 ÷ 7✓ Volume = 98✓cm^3	1M 1CA (2)
2.2.5	Area = 7,2 × 7,2✓ + 4✓(7,2 × 14,2) ✓ Area = 51,84 + 408,96 Area = 460,8 ✓cm^2 Cost = 460,8✓ × R0,05✓ Cost = R23,04✓ Or Cost = 460,8 × 5✓ = 2304 c Cost = R23,04✓	3MA Area 1A (460,8) 1SF 1A R23,04 with correct conversion 1SF 1C R23,04 with correct conversion (6)
		[30]

QUESTION 3 [30] *Do not penalise if 'R' is omitted.		
Question	**Solution**	**Explanation**
3.1 3.1.1	(a) Any Time Per Second = R1,20 ÷ 60 = R0,02✓ = 2✓ c Or Any Time Per Second = 120✓ ÷ 60 = 2 ✓c	 1MA 1C 1C 1A (2)
	(b) Pay Per Second = R0,79 ÷ 60 = R0,013✓ = 1,3✓ c (accept 1 cent) Or Pay Per Second = 79 ✓÷ 60 = 1,3✓ c	 1MA 1C 1C 1A (2)
3.1.2	A = R1,20✓ × 1,5✓ = R1,80✓ **Or** A = R1,20 + R0,60✓ = R1,80✓ (**answer only full marks**)	1RT 1M × 1,5 1A 1RT 1M + R0,60 1A (3)
3.1.3	B = R0,75 ✓× 0,5✓ = R0,38✓ **Or** B = R1,13✓ – R0,75✓ = R0,38✓ (**answer only full marks**)	1RT 1M × 1,5 1A 1RT 1M + R0,60 1A (3)
3.1.4	Total call time = R12,00 ÷ R1,20✓ = 10 min ✓ **(answer only full marks)**	1M 1A (2)
3.1.5	60 sec : R0,66 50 sec : Cost Cost = 50 × R0,66 ÷ 60✓	 1M

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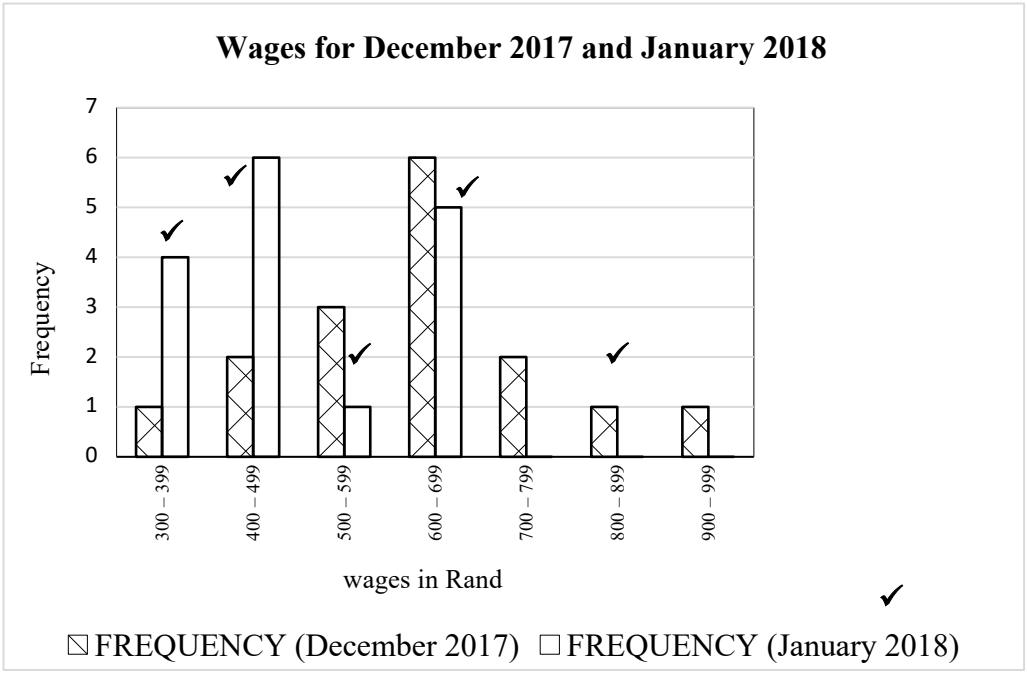
	Cost = R0,55✓**]** **(answer only full marks)**	1A (2)
3.1.6	<u>Pay Per Second</u> Cost = 120 × R0,79 ÷ 60✓ or 2 × R0,79 Cost = R1,58✓ **(answer only 2 marks)** <u>Sim–Sonke</u> Cost = 120 × R0,75 ÷ 60✓ or 2 × R0,75 Cost = R1,50✓ **(answer only full marks) (award 1 mark for R1,5)** Difference = R1,58 – R1,50 Difference = ±R0,08✓ **(accept R0,08 or – R0,08 or 8c or – 8c)**	 1M 1A 1M 1A 1P incorrect rounding of money 1MA (5)
3.2 3.2.1	Budget Bank charge = R10✓	1RT (1)
3.2.2	ABS Bank charge = R4,20 + R1,25 × (R1 525 ÷ 100✓) ✓ = R23,26✓	1RT and 1M ÷ 100 1A (3)
3.2.3	ABS Bank charge = R5,10 + R2,50 × (R1 400 ÷ 100) ✓ = R40,10✓ BUDGET Bank charge = R5 + R2 × (R1 400 ÷ 100) = R33✓ Therefore, BUDGET Bank✓ charges a cheaper fee Or ABS Bank charge = R5,10 + R2,50 × 14✓ = R40,10✓ BUDGET Bank charge = R5 + R2 × 14 = R33✓ Therefore, BUDGET Bank✓ charges a cheaper fee	1M 1A 1A 1A 1M 1A 1A 1A (4)
3.2.4	ABS Bank charge = 0 + 3 × 5 = R15✓ BUDGET Bank charge = 4× x R2,50 = R10✓ Therefore, BUDGET Bank✓ charges a cheaper fee	1MA 1MA 1A (3)
		[30]

QUESTION 4 [30] *Do not penalise if units are omitted.		
Question	**Solution**	**Explanation**
4.1 4.1.1	Basic wage per day = R1 000 ÷ 20 ✓ = R50✓ **(answer only full marks)**	1M 1 A (2)
4.1.2	Wage for one day = R50 + 50 × 4✓ = R250✓ **(answer only full marks)**	1M 1A (2)
4.1.3	R700 – R50 = R650✓ R650 ÷ 50✓ = 13✓ customers for the day **(answer only full marks)**	1MA Subtract daily wage 1MA Divide by 50 1 A (3)
4.2 4.2.1	A = R7 000✓✓	2 MA (2)
4.2.2	Total wage per month = R1 000 ✓+ R50 ✓× number of customers✓ Or Total wage per month = R1 000 ✓+ R50✓ × n ✓ Or Total wage per month = R1 000 ✓+ R50 ✓× *x*✓	3A (3)

4.2.3	 Marks: 1 appropriate title 4 correctly plotting first 4 points 1 CA plotting (120; 7 000) 1 correct straight line	(7)
4.2.4	(a) Number of customers✓ (b) Direct✓ proportion (c) Increasing✓ graph	3A (3)
4.3 4.3.1	Relative bus fare per trip = R356,40✓ ÷ 40✓ = R8,91✓ **(answer only full marks)**	2M 1A (3)
4.3.2	As the number of bus trips increases✓, the relative bus fare per trip decreases✓	2E (2)
4.3.3	Total cash bus fare for single trips = R10,80 × 44✓ = R475,20✓ Amount saved = R475,20 – R356,40 = R118,80✓	1SF 1A 1MA (3)
		[30]

QUESTION 5 [30] *Do not penalise if 'R' is omitted.		
Question	**Solution**	**Explanation**

5.1	Mode = R615✓✓	2RT	(2)
5.2	Range = 900 – 399✓ = R501✓ Full marks for answer only **(answer only full marks)**	1RT 1A	(2)
5.3 5.3.1	Mean = $\frac{\text{R10 140}✓}{16✓}$ = R633,75	2MA	(2)
5.3.2	6✓✓	2RT	(2)
5.3.3	Percentage = $\frac{10✓}{16} \times 100\%✓$ = 62,5✓%	1A - 10 1M 1A	(3)
5.3.4	Mean wage December 2017 = R633,75 Mean wage January 2018 = $\frac{\text{R7 892}✓}{16}$ = R493,25✓ Difference = R633,75 – R493,25✓ Difference = R 140,5✓	1A (R7 892) and 1A (R493,25) 1M (subtracting) 1CA	(4)

5.4	374 385 394 399 421 425 450 **450** **455** 490 550 601 602 605 641 650✓ Median = $\frac{450+455}{2}$ ✓ = R452,50✓ (no mark for R452,5)	1A sort 1M 1 A (3)
5.5	**Wages earned in December 2017 and January 2018** **ITEM (in rand)** / **FREQUENCY (December 2017)** / **FREQUENCY (January 2018)** 300 – 399: 1✓ / 4✓ 400 – 499: 2✓ / 6✓ 500 – 599: **3** / **1** 600 – 699: 6✓ / 5✓ 700 – 799: 2 / 0 800 – 899: 1 / 0 900 – 999: 1 / 0	6A (accurate reading and adding from tables) (6)
5.6	 Marks: 1 Legend 4 correctly plotting first 4 double bars 1 correctly plotting the 3 single bars	(6)
		[30]
		TOTAL: 150

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